Reinforcement Learning & Bayesian Methods for Cryptocurrency Portfolio Management

DQN · REINFORCE · Market Simulation · Bayesian Bandits

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ORIE5570: Reinforcement Learning with Operations Research Applications

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Slide 1 Introduction & Professional Background

Cornell University · ORIE · ORIE5570

About Me

Graduate Student, Cornell University

MS CIS
MS Applied Information Science
Btech Computer Engineering

Research interests:

- Reinforcement learning & sequential decision-making
- Optimization Methods, numerical methods
- ML Hardware, ML Engg, Distr. Systems
- Financial optimization & quantitative modeling
- Bayesian inference & uncertainty quantification

This Project

Course: ORIE5570 — RL with OR Applications
4-part end-to-end study: RL & Bayesian methods on live crypto markets

Asset universe: BTCB, ETH, SOL, BNB, USDC (2024–2026)

Combines trading policy design with systemic / regulatory analysis

Four-Part Framework

Part 1	Environment design & 13-dim feature engineering
Part 2	Deep Q-Network (DQN) agent — off-policy trading
Part 3	REINFORCE + 3 trader archetypes; Shapley attribution; 7-composition market simulation
Part 4	Bayesian bandits: UCB1, Bayes-UCB, Thompson Sampling on 5 crypto arms

Why Crypto?

Extreme regime shifts, manipulative microstructure, and herd dynamics make crypto an ideal stress-test for adaptive RL policies and Bayesian sequential decision-making.

Slide 2 Benefits to the End User

BlackRock-like Asset Manager · Regulatory / Government Body (DOB)

Institutional Asset Manager (BlackRock-like)

▸ **Capital Preservation**

-6% return vs -46% buy-and-hold during bear market;
preserved 94% of capital

▸ **Explainable AI**

Shapley values identify exact signals per strategy
(RSI for arb · volume for manip · 7-day return for herder)
Auditable to compliance teams

▸ **Regime-Aware Allocation**

Policies adapt dynamically to rise / steady / decline

▸ **Bayesian Portfolio Selection**

Thompson Sampling allocates across 5 assets with
quantified posterior uncertainty; $\Gamma_T = 0.149$

★ *The same model serves both trading desks and regulatory surveillance simultaneously.*

Regulatory / Government Body (DOB)

▸ **Manipulator Detection**

Front-runners identified via microstructure signals
(MA deviation, realized vol, volume) — not price level;
requires order-book surveillance, not price triggers

▸ **Crisis Early Warning**

Herder markets produce 6.75× more cascade events
than arbitrageur markets (0.054 vs 0.008)

▸ **Stablecoin Surveillance**

Arbitrageur surge predicts USDT peg deviation;
grounded in actual USDT-USD hourly peg data

▸ **Regulatory Sweet Spot**

Equal mix (33/33/34) achieves best balance of
crisis frequency, stablecoin stability, efficiency

Slide 3 Data Description

17,248 hourly observations · 5 crypto assets · Mar 2024 – Feb 2026

Primary Dataset — BTCB-USD

Asset:	BTCB-USD (Bitcoin BEP-20, Yahoo Finance)
Frequency:	Hourly OHLCV
Observations:	17,248 hourly rows
Full period:	March 2024 – February 2026
Train split:	March 2024 – October 2025
Test split:	October 2025 – February 2026
State vector:	13-dim: RSI, MACD, vol, MA crossovers, regime

Supplementary Data

Cross-assets:	ETH, Gold, Silver, S&P500 (correlation by regime)
Stablecoin:	USDT-USD hourly peg data (3,394 hrs; mean dev = 0.058%)
Bandit arms:	ETH, SOL, BNB, USDC via yfinance; resampled to 8h
Aligned obs.:	1,990 eight-hour periods × 5 arms

Key Periods

Training (Mar 2024 – Oct 2025)
~13,000 hourly obs.; varied bull/bear regimes

Test (Oct 2025 – Feb 2026)
BTCB declined 46.2% ← severe bear market
Critical stress-test period for all agents

Bandit horizon T = 500
100 bootstrap replications

Market Regimes & Cross-Asset Correlations

Regime	% Hours	ETH Corr	Gold Corr
Steady	64%	0.756	+0.089
Decline	21%	0.810 ↑	+0.262 ↑
Rise	15%	0.677	−0.122 ↓

Gold goes negative in rising markets — flight-to-safety reversal

Slide 4 Methodology

RL Agents (Parts 1–3) · Bayesian Bandits (Part 4) · Market Simulation (Part 3)

RL Trading Agents (Parts 2 & 3)

DQN — Part 2

Off-policy Q-learning · ε-greedy exploration · Experience replay buffer
Actions: {Buy, Hold, Sell} · 13-dim state vector · Reward: clipped log-return

REINFORCE + Baseline — Part 3

On-policy Monte Carlo PG · 2-layer MLP (128 units) · Adam lr=3e-4
Update: $\nabla J = E[\log \pi(a|s;\theta) \cdot (G_t - V(s))]$

Three reward-shaped trader archetypes:

Arbitrageur: $r - 0.4 \cdot \sigma_{24} \cdot f\_crypto$ (variance penalty)
Manipulator: $r + 0.15 \cdot \nabla \cdot \mathrm{sign}(r_4 h) \cdot d$ (front-running)
Herder: $r + 0.25 \cdot \mathrm{sign}(r_{24} h) \cdot d$ (momentum chasing)

Explainability — SHAP KernelExplainer

Applied to Buy-action probability $\pi(\mathrm{Buy}|s;\theta)$
200 test states · 80 coalition samples · Mean |SHAP| reported

Bayesian Bandits (Part 4)

K=5 arms (BTC, ETH, SOL, BNB, USDC)
Reward: Y_t = 1{8h log-return > 0} (Bernoulli)
Prior: Beta(3,3) uninformative · Horizon T=500 · 100 replications

UCB1	$\hat{\mu}\_k + \sqrt{2 \ln t / n\_k}$	frequentist bonus
Bayes-UCB	$Q(1-1/t, \mathrm{Beta}(\alpha\_k,\beta\_k))$	posterior quantile
Thompson	sample $\tilde{\theta}\_k \sim \mathrm{Beta}(\alpha\_k,\beta\_k)$	posterior sample
Greedy	$\mathrm{argmax}\_k\ \alpha\_k/(\alpha\_k+\beta\_k)$	posterior mean

Info ratio: $\Gamma\_t = [E[\Delta\_t]]^2 / I\_t(\tilde{\theta}; A\_t)$ via Monte Carlo (N_mc=150)

Market Simulation (Part 3)

7 trader compositions · 500 test-set steps

Aggregate signal: S_t = w_arb·d_arb + w_man·d_man + w_herd·d_herd

Metrics:
- Crisis frequency (decline regime + |S_t|>0.3)
- Stablecoin stability (1 / mean USDT peg deviation)
- Market efficiency (1 − |lag-1 autocorrelation|)

Slide 5 Key Findings

Trading Performance · Explainability · Systemic Risk · Bayesian Bandits

Trading Performance

- REINFORCE-Manipulator: −6% vs −46% buy-and-hold
  → preserved 94% of capital during severe bear market

- DQN collapsed to constant HOLD
  (Q-values: Hold=0.0043, Buy=0.0041, Sell=0.0033)
  Stochastic RL adapts; deterministic RL freezes

- Bayesian: Thompson Sampling regret = 9.19 ± 0.78
  Optimal arm: BNB (θ^* = 0.519); all arms within 1.5 pp
  Info ratio Γ_T = 0.149 ≪ 0.5 theoretical bound

Shapley Value Attribution

Arbitrageur	RSI + 24h return (mean-reversion signals)
Manipulator	MA deviation + realized vol + volume (pure microstructure — zero trend signals)
Herder	7-day return dominant; blind to RSI and volume (follows trend, ignores contrarian signals)

Systemic & Regulatory Findings

- Herder-dominated markets: 6.75× more crisis events than arbitrageur markets (0.054 vs 0.008)

- All-manipulator markets: 0 measured crises
  — but wealth extraction is silent
  → Requires order-book surveillance, NOT price monitoring

- Arbitrageur surge → lowest stablecoin stability (18.1) vs all-manipulator (22.6); predicts USDT peg stress

- Equal mix (33/33/34) achieves best regulatory profile: balanced crisis frequency, stability & market efficiency

Core Takeaway

Stochastic RL dominates deterministic Q-learning in volatile markets. Manipulator detection requires microstructure surveillance. Trader diversity minimizes systemic risk.